

NYPD Shooting Incidents

Data Sets for Importing,
Tidy & Transforming, and
Visualizing Data



Step 1: Data Importing

```
library(tidyverse)
```

```
library(tinytex)
```

```
library(lubridate)
```

```
url <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD"
```

```
nypd <- read_csv(url)
```



Step 2: Tidy & Transform Data

```
summary(nypd)
```

Removing Unwanted Columns from the Data Set

```
nypd = nypd %>%  
  select(-c(INCIDENT_KEY, OCCUR_TIME, PRECINCT, JURISDICTION_CODE,  
            STATISTICAL_MURDER_FLAG, PERP_RACE, VIC_RACE,  
            X_COORD_CD, Y_COORD_CD, Latitude, Longitude, Lon_Lat))
```

Changing the “OCCUR_DATE” column data type into date

```
class(nypd$OCCUR_DATE)
```

```
[1] "character"
```

```
nypd <- nypd %>% mutate(OCCUR_DATE = mdy(OCCUR_DATE))
```

```
class(nypd$OCCUR_DATE)
```

```
[1] "Date"
```

changing names for some columns

```
colnames(nypd)[1] <- "Date"
```

```
colnames(nypd)[2] <- "Borough"
```

```
colnames(nypd)[3] <- "place"
```

```
colnames(nypd)[4] <- "Perp_Age"
```

```
colnames(nypd)[5] <- "Perp_Gender"
```

```
colnames(nypd)[6] <- "Vic_Age"
```

```
colnames(nypd)[7] <- "Vic_Gender"
```

Step 3: Visual Data

Step 3: Visualizing & Analyzing

Creating frequency table for victim age

```
bn <- table(nypd$Vic_Age)
```

Creating barplot for victim age

```
barplot(  
  bn,  
  names = bVic_Age,  
  xlab = "Victim age",  
  ylab = "Shootings",  
  main = "Incidents By Victim Age",  
  col = "green"  
)
```

Step 3: Visualizing & Analyzing

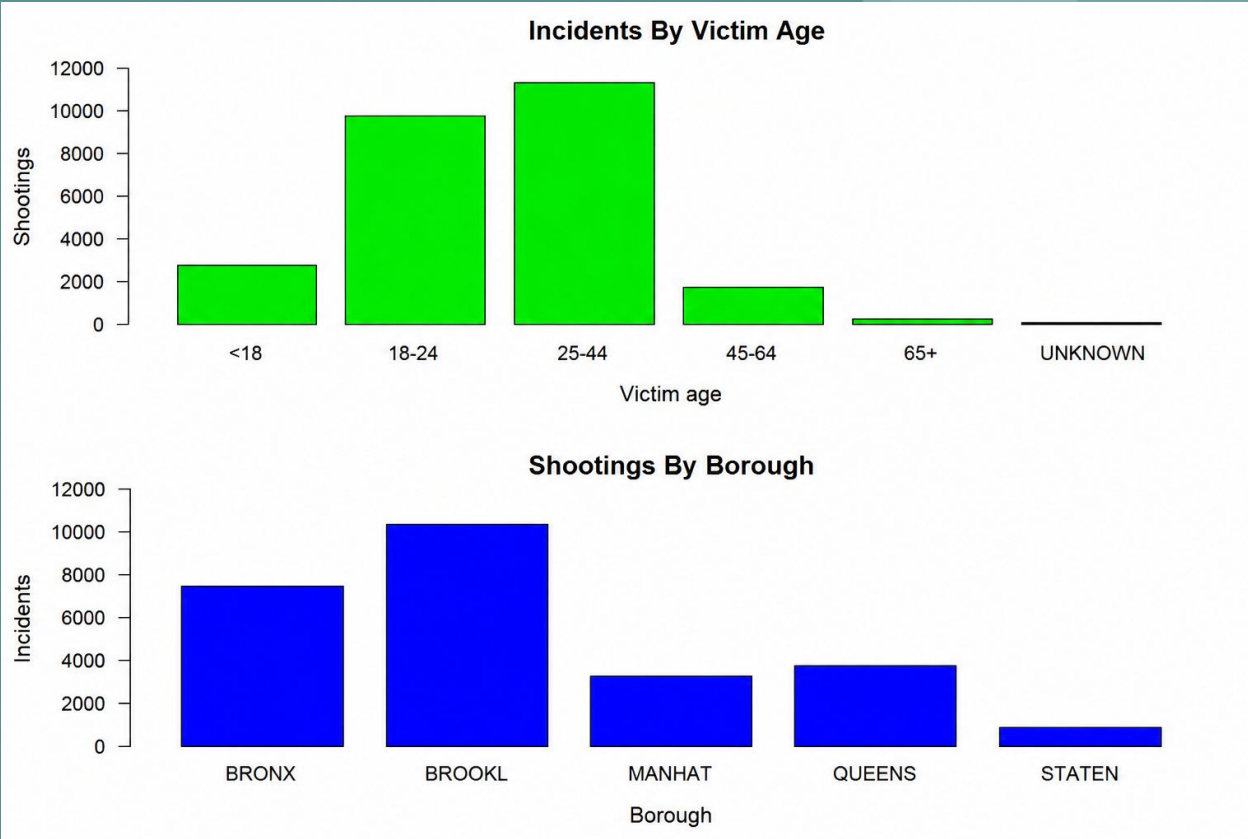
Creating frequency table for boroughs

```
an <- table(nypd$Borough)
```

Creating barplot for boroughs

```
barplot(  
  an,  
  names = aBorough,  
  xlab = "Borough",  
  ylab = "Incidents",  
  main = "Shootings By Borough",  
  col = "blue"  
)
```

Step 3: Visual Data



Step 4: Bias Identification

Age Distribution Bias:

The majority of shooting incidents involve victims between ages **18–44**, which may reflect higher exposure to public environments, nightlife, or socioeconomic factors. However, this could also create a bias where other age groups appear less affected simply because they represent smaller portions of the population.

